

On the irrationality of infinitely many odd values of the zeta function

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1 Introduction

It is widely known that the values that takes the Riemann zeta function at even positive integers are irrational. More precisely, the values $1, \zeta(2), \zeta(4), \dots, \zeta(2k), \dots$ are linearly independent over $\mathbb{Q}$ and each of those numbers is transcendental. This follows from the formula

$$\zeta(2k) = (-1)^{k-1} 2^{2k-1} B_{2k} \frac{\pi^{2k}}{(2k)!}$$

for $k \geq 1$, where $B_{2k} \in \mathbb{Q}$ is the $2k$ -th Bernoulli number.

However, we know very little about the values of ζ at odd positive integers. To this day, no one has found an explicit formula for computing $\zeta(2k+1)$. We don't even know if these values are irrational or not, with some few exceptions. Indeed, Apéry showed in [1] that $\zeta(3)$ is irrational. Also, Gutnik showed in [3] that for every $q \in \mathbb{Q}$, at least one of the following numbers:

$$3\zeta(3) + q\zeta(2) \quad \text{or} \quad \zeta(2) + 2q \log(2)$$

is irrational.

The goal of this document is to explain in more detail a major result proven in [2]. In it, Ball and Rivoal show that the values $1, \zeta(3), \zeta(5), \dots, \zeta(2k+1), \dots$ span a $\mathbb{Q}$ -vector space of infinite dimension, which in particular implies that there are infinitely many $k \geq 1$ such that $\zeta(2k+1)$ is irrational. To prove this result, we are going to use Nesterenko's independence criterion to bound from below the dimension of the $\mathbb{Q}$ -vector space spanned by

$$\{1, \zeta(3), \zeta(5), \dots, \zeta(2k+1)\}.$$

More precisely, we are going to show the next result:

Theorem 1

For $a \geq 3$ an odd integer, if we define $\delta(a) = \dim_{\mathbb{Q}} \text{Vect}_{\mathbb{Q}}\{1, \zeta(3), \dots, \zeta(a)\}$, then

$$\delta(a) \geq \frac{1}{3} \log(a).$$

Additionally, for every $\varepsilon > 0$, there is some $A(\varepsilon) \in \mathbb{N}$ such that if $a > A(\varepsilon)$, then

$$\delta(a) \geq \frac{1 - \varepsilon}{1 + \log(2)} \log(a).$$

We note that $\frac{1}{1 + \log 2} \approx 0,5906$.

We will also show that

Theorem 2

There exists an odd integer $j \leq 169$ such that $1, \zeta(3), \zeta(j)$ are linearly independent over $\mathbb{Q}$.

This is in itself an interesting result, but we will use it to show [Theorem 1](#). We will also show Apéry's theorem, which tells us that $\zeta(3)$ is irrational. This translates into the fact that $\delta(3) = 2$.

The main result for showing our theorem will be Nesterenko's linear independence criterion, which we will admit:

Lemma 1 (Nesterenko)

Let $N \geq 2$ be an integer and $\theta_1, \dots, \theta_N$ real numbers such that there exist N sequences $(p_{1,n})_{n \geq 0}, \dots, (p_{N,n})_{n \geq 0}$ such that

- For every $k \in \llbracket 1, N \rrbracket$ and $n \in \mathbb{N}$ we have $p_{k,n} \in \mathbb{Z}$,
- There exists $\alpha \in \mathbb{R}_+^*$ such that

$$\left| \sum_{k=1}^N p_{k,n} \theta_k \right| = \alpha^{n+o(n)}.$$

- There exists $\beta > 1$ such that for every $k \in \llbracket 1, N \rrbracket$

$$|p_{k,n}| \leq \beta^{n+o(n)}.$$

Under these conditions,

$$\dim_{\mathbb{Q}} \text{Vect}_{\mathbb{Q}}\{\theta_1, \dots, \theta_N\} \geq \frac{\log \beta - \log \alpha}{\log \beta}.$$

If we fix $a \in \mathbb{N}$ odd with $a \geq 3$, the proof consists in constructing the $N = \frac{a+1}{2}$ sequences of Nesterenko's independence criterion for the real numbers $1, \zeta(3), \dots, \zeta(a)$.

2 The definitions

Here we define the main tools of the proof. Since we are going to use a lot of derivatives, we define the operator $D_\lambda = \frac{1}{\lambda!} \frac{d^\lambda}{dz^\lambda}$, that is, $D_\lambda f|_{z=z_0} = \frac{1}{\lambda!} f^{(\lambda)}(z_0)$. For $\alpha \in \mathbb{R}$, we will use the Pochhammer symbol which is defined as

$$(\alpha)_0 = 1 \text{ and } (\alpha)_k = \alpha(\alpha+1) \cdots (\alpha+k-1) \text{ for } k \geq 1.$$

We let $\log$ denote the natural base logarithm and $\log_p$ denote the base p logarithm, where p is a prime. We fix, throughout this document $a, r \in \mathbb{N}$ such that $1 \leq r < \frac{a}{2}$, and such that a is odd. Note that this implies that $a \geq 3$. Let $n \in \mathbb{N}$. We consider the meromorphic function

$$R_n(z) = n!^{a-2r} \frac{(z - rn + 1)_{rn} (z + n + 2)_{rn}}{(z + 1)_{n+1}^a}$$

and the series

$$S_n(z) = \sum_{k=0}^{\infty} \frac{R_n(k)}{z^k}.$$

Since

$$\deg(R_n) = 2rn - a(n+1) = n(2r - a) - a \leq -a - 1 \leq -2$$

there are $M, C > 0$ such that for every $z \in \mathbb{C}$ with $|z| > M$,

$$|R_n(z)| < \frac{C}{|z|^2}.$$

This implies that for $z \in \mathbb{C}$ with $|z| \geq 1$:

$$\begin{aligned} |S_n(z)| &= \left| \sum_{k=0}^{\infty} \frac{R_n(k)}{z^k} \right| \\ &\leq \sum_{k=0}^M \left| \frac{R_n(k)}{z^k} \right| + \sum_{k>M} \left| \frac{R_n(k)}{z^k} \right| \\ &\leq \sum_{k=0}^M |R_n(k)| + \sum_{k>M} |R_n(k)| \\ &\leq \sum_{k=0}^M |R_n(k)| + \sum_{k>M} \frac{C}{|k|^2} \\ &< +\infty \end{aligned}$$

For $l \in \llbracket 1, a \rrbracket$ and $j \in \llbracket 0, n \rrbracket$ we note

$$c_{l,j,n} = D_{a-l}(R_n(z)(z+j+1)^a)|_{z=-(j+1)}$$

Since $R_n(X)(X+j+1)^a \in \mathbb{Q}(X)$, we observe that $c_{l,j,n} \in \mathbb{Q}$. We similarly define the polynomials

$$P_{0,n}(z) = \sum_{l=1}^a \sum_{j=1}^n \sum_{k=0}^{j-1} \frac{-c_{l,j,n}}{(k+1)^l} z^{j-k}$$

$$P_{l,n}(z) = \sum_{j=0}^n c_{l,j,n} z^j \quad \text{for } l \in \llbracket 1, a \rrbracket$$

We note that for $l \in \llbracket 0, a \rrbracket$, we have $P_{l,n} \in \mathbb{Q}[X]$.

We also consider the polynomial

$$Q_{r,a}(X) = rX^{a+2} - (r+1)X^{a+1} + (r+1)X - r.$$

We will prove that this polynomial has a unique root $s_0 \in (0, 1)$. We compute:

$$Q'_{r,a}(X) = (a+2)rX^{a+1} - (a+1)(r+1)X^a + r+1$$

$$\begin{aligned} Q''_{r,a}(X) &= (a+1)(a+2)rX^a - a(a+1)(r+1)X^{a-1} \\ &= (a+1)X^{a-1}((a+2)rX - a(r+1)) \end{aligned}$$

Over this interval, $Q''_{r,a}$ has the same sign as $(a+2)rX - a(r+1)$, which is a strictly increasing affine function which is equal to $2r - a < 0$ in 1. Therefore, $Q''_{r,a} < 0$ and $Q'_{r,a}$ is strictly decreasing. We compute

$$Q'_{r,a}(0) = r+1 > 0$$

$$Q'_{r,a}(1) = 2r - a < 0$$

If we let s_1 the unique root of $Q'_{r,a}$ in $[0, 1]$, which exists by the intermediate value theorem, then $Q_{r,a}$ is strictly increasing before s_1 , and strictly decreasing after s_1 , so that it attains a maximum ε_+ in s_1 . Since $Q_{r,a}(1) = 0$, then $\varepsilon_+ > 0$. We similarly note

$$\varepsilon_- = Q_{r,a}\left(\frac{r}{r+1}\right) = -\left(\frac{r}{r+1}\right)^{a+1} \frac{2r+1}{r+1} < 0.$$

By the intermediate value theorem applied to $Q_{r,a}$ on $(\frac{r}{r+1}, 1)$, there is a unique $s_0 \in (\frac{r}{r+1}, 1)$ such that $Q(s_0) = 0$, and we easily observe that this is the unique root of Q in $(0, 1)$.

Finally, we define

$$\varphi_{r,a} = ((r+1)s_0 - r)^r((r+1) - rs_0)^{r+1}(1 - s_0)^{a-2r}.$$

This is an expression which will appear later in our computations. It is important to note that we can approximate the value of s_0 with a computer, which in turn will give us an approximation of $\varphi_{r,a}$.

We will use the convention that, for an integer $n \geq 1$:

$$\text{Li}_n(z) = \sum_{k=0}^{+\infty} \frac{z^k}{(k+1)^n}.$$

so that $\text{Li}_1(z) = -\frac{\log(1-z)}{z}$ diverges near 1 and $\text{Li}_n(1) = \zeta(n)$ for $n \geq 2$.

3 Sketch of the proof

In this section we give a general overview of the proof. If we let $d_{2n} = \text{lcm}(1, \dots, 2n)$, then we will define

$$p_{0,n} = d_{2n}^a P_{0,2n}(1)$$

$$p_{k,n} = d_{2n}^a P_{2k+1,2n}(1) \quad \text{for } k \in \llbracket 1, \dots, (a-1)/2 \rrbracket.$$

We will show that $p_{k,n} \in \mathbb{Z}$ for every $k \in \llbracket 0, \frac{a-1}{2} \rrbracket$, and if we let $\ell_n = d_{2n}^a S_{2n}(1)$, we will show that

$$\ell_n = p_{0,n} + \sum_{k=1}^{\frac{a-1}{2}} p_{k,n} \zeta(2k+1).$$

With good choices for α and β (in terms of a and r) we will prove

$$|\ell_n| = \alpha^{n+o(n)}$$

$$|p_{k,n}| \leq \beta^{n+o(n)}$$

and so we will apply Nesterenko independence criterion to obtain

$$\delta(a) \geq \frac{\log \beta - \log \alpha}{\log \beta}.$$

After simplification, this expression will be equal to

$$\delta(a) \geq \frac{(a-2r) \log 2 + (2r+1) \log(2r+1) - \log(\varphi_{r,a})}{a + (a-2r) \log(2) + (2r+1) \log(2r+1)}$$

After this, we will choose a specific r to obtain [Theorem 2](#) and then [Theorem 1](#).

4 Main lemmas

In this section we prove the lemmas we will use to prove our two theorems:

Lemma 2

For $n \in \mathbb{N}$ we have:

$$S_n(1) = P_{0,n}(1) + \sum_{l=2}^a P_{l,n}(1) \zeta(l) \quad (1)$$

Also, for $l \in \llbracket 1, a \rrbracket$, if $(n+1)a + l$ is odd, then $P_{l,n}(1) = 0$. In particular, if n and l are even, then $P_{l,n}(1) = 0$. So, $S_n(1)$ is a linear combination of the odd ζ values:

$$S_n(1) = P_{0,n}(1) + \sum_{k=1}^{\frac{a-1}{2}} P_{2k+1,n}(1) \zeta(2k+1). \quad (2)$$

Proof. We apply the partial fraction decomposition theorem on $R_n(z)$. Its denominator is equal to $\prod_{j=0}^n (z+1+j)^a$ and $\deg(R_n) \leq -2 < 0$, so there exist $(b_{l,j,n})_{(l,j)} \in \mathbb{Q}^{a(n+1)}$ such that

$$R_n(z) = \sum_{l=1}^a \sum_{j=0}^n \frac{b_{l,j,n}}{(z+j+1)^l}.$$

Let $(l, j) \in \llbracket 1, a \rrbracket \times \llbracket 0, n \rrbracket$. There exists Q_n an holomorphic function over $B_{\mathbb{C}}(-(j+1), \frac{1}{2})$ such that over this open neighborhood :

$$R_n(z)(z+j+1)^{l-1} = Q_n(z) + b_{1,j,n}(z+j+1)^{l-2} + \dots + \frac{b_{l,j,n}}{z+j+1} + \dots + \frac{b_{a,j,n}}{(z+j+1)^{a-l+1}}.$$

This function has a pole on $z = -(j+1)$ with residue $b_{l,j,n}$ of order $a-l+1$. We can now show that $b_{l,j,n} = c_{l,j,n}$ by extracting the residue:

$$\begin{aligned} b_{l,j,n} &= D_{(a-l+1)-1}(R_n(z)(z+j+1)^{l-1}(z+j+1)^{a-l+1})|_{z=-(j+1)} \\ &= D_{a-l}(R_n(z)(z+j+1)^a)|_{z=-(j+1)} \\ &= c_{l,j,n}. \end{aligned}$$

We thus obtain the next equality:

$$R_n(z) = \sum_{l=1}^a \sum_{j=0}^n \frac{c_{l,j,n}}{(z+j+1)^l}.$$

By definition of S_n and by the previous calculations, we have

$$\begin{aligned} S_n(z) &= \sum_{k=0}^{\infty} \sum_{l=1}^a \sum_{j=0}^n \frac{1}{z^k} \frac{c_{l,j,n}}{(k+j+1)^l} \\ &= \sum_{l=1}^a \sum_{j=0}^n c_{l,j,n} z^j \left(\sum_{k=0}^{\infty} \frac{1}{z^k} \frac{1}{(k+1)^l} - \sum_{k=0}^{j-1} \frac{1}{z^k} \frac{1}{(k+1)^l} \right) \\ &= \sum_{l=1}^a \sum_{j=0}^n c_{l,j,n} z^j \operatorname{Li}_l(1/z) - \sum_{l=1}^a \sum_{j=1}^n c_{l,j,n} \sum_{k=0}^{j-1} \frac{1}{(k+1)^l} z^{j-k} \\ &= \sum_{l=1}^a P_{l,n}(z) \operatorname{Li}_l(1/z) + P_{0,n}(z). \end{aligned}$$

Let us show that

$$\lim_{\substack{z \rightarrow 1 \\ |z| > 1}} P_{1,n}(z) \operatorname{Li}_1(1/z) = 0.$$

Recall that there existed $M, C > 0$ such that for $z \in \mathbb{C}$, if $|z| > M$, then

$$|R_n(z)| \leq \frac{C}{|z|^2}.$$

We have

$$\begin{aligned}
P_{1,n}(1) &= \sum_{j=0}^n c_{1,j,n} \\
&= \sum_{j=0}^n D_{a-1} (R_n(z)(z+j+1)^a)_{|z=-(j+1)} \\
&= \sum_{j=0}^n \text{Res} (R_n, -(j+1))
\end{aligned}$$

Let $L > \max(n+1, M)$. The residue theorem with $\gamma(t) = Le^{it}$ gives us

$$P_{1,n}(1) = \frac{1}{2\pi i} \int_0^{2\pi} L e^{it} R_n (L e^{it}) dt.$$

By the triangle inequality, we have

$$|P_{1,n}(1)| \leq \frac{1}{2\pi} \int_0^{2\pi} L \cdot \frac{C}{|L e^{it}|^2} dt = \frac{C}{L}$$

Therefore $P_{1,n}(1) = 0$, and thus there exists $Q \in \mathbb{Q}[X]$ such that for $z \in \mathbb{C}$,

$$P_{1,n}(z) = Q(z)(z-1).$$

For every $z \in \mathbb{C} \setminus B_{\mathbb{C}}(0, 1)$, we have:

$$\begin{aligned}
|P_{1,n}(z) \text{Li}_1(1/z)| &= \left| Q(z)(z-1)z \log \left(1 - \frac{1}{z} \right) \right| \\
&= \left| Q(z) \cdot z^2 \cdot \left(1 - \frac{1}{z} \right) \cdot \log \left(1 - \frac{1}{z} \right) \right|.
\end{aligned}$$

When $z \rightarrow 1$, we have $(1 - 1/z) \log(1 - 1/z) \rightarrow 0$, and thus we can conclude

$$\lim_{\substack{z \rightarrow 1 \\ |z| > 1}} P_{1,n}(z) \text{Li}_1(1/z) = 0.$$

Also, $\text{Li}_l(1) = \zeta(l)$ for $l \geq 2$, and so we obtain:

$$S_n(1) = P_{0,n}(1) + \sum_{l=2}^a P_{l,n}(1) \zeta(l).$$

Let $l \in \llbracket 2, a \rrbracket$. It remains to verify that if $(n+1)a + l$ is odd, then $P_{l,n}(1) = 0$. In order to do that, for every $j \in \llbracket 0, n \rrbracket$ we let:

$$\Phi_{n,j}(z) = R_n(-z-1)(j-z)^a$$

So, for every $j \in \llbracket 0, n \rrbracket$ we have

$$c_{l,j,n} = (-1)^{a-l} D_{a-l}(\Phi_{n,j}(z))|_{z=j}$$

since, using that $(x \mapsto -x - 1)$ is an affine function:

$$\begin{aligned} c_{l,j,n} &= D_{a-l}(R_n(z)(z + j + 1)^a)|_{z=-(j+1)} \\ &= D_{a-l}(\Phi_{n,j}(-z - 1))|_{z=-(j+1)} \\ &= D_{a-l}(\Phi_{n,j}(z))|_{z=j} \cdot (-1)^{a-l}. \end{aligned}$$

Furthermore, we remark that for every $\alpha \in \mathbb{R}$ and every $k \in \mathbb{N}$,

$$\begin{aligned} (\alpha)_k &= \prod_{i=0}^{k-1} (\alpha + i) \\ &= (-1)^k \prod_{i=0}^{k-1} (-\alpha - i) \\ &= (-1)^k \prod_{i=0}^{k-1} (-\alpha - (k-1) + i) \\ &= (-1)^k (-\alpha - k + 1)_k. \end{aligned}$$

Let $j \in \llbracket 0, n \rrbracket$. Using the previous identity, we obtain:

$$\begin{aligned} \Phi_{n,n-j}(n-z) &= R_n(-(n-z)-1)(n-j-(n-z))^a \\ &= n!^{a-2r} \frac{(z-(r+1)n)_{rn}(z+1)_{rn}}{(z-n)_{n+1}^a} (z-j)^a \\ &= n!^{a-2r} \frac{(-1)^{rn}(-z+n+1)_{rn}(-1)^{rn}(-z-rn)_{rn}}{(-1)^{(n+1)a}(-z)_{n+1}^a} (-1)^a (j-z)^a \\ &= (-1)^{na} R_n(-z-1)(j-z)^a \\ &= (-1)^{na} \Phi_{n,j}(z) \end{aligned}$$

Finally, we compute:

$$\begin{aligned}
P_{l,n}(1) &= \sum_{j=0}^n c_{l,j,n} \\
&= \sum_{j=0}^n D_{a-l}(\Phi_{n,j}(z))|_{z=j} (-1)^{a-l} \\
&= \sum_{j=0}^n D_{a-l}((-1)^{na} \Phi_{n,n-j}(n-z))|_{z=j} (-1)^{a-l} \\
&= \sum_{j=0}^n (-1)^{na} (-1)^{a-l} D_{a-l}(\Phi_{n,n-j}(z))|_{z=n-j} (-1)^{a-l} \\
&= \sum_{j=0}^n (-1)^{na} (-1)^{a-l} c_{l,n-j,n} \\
&= (-1)^{(n+1)a+l} P_{l,n}(1)
\end{aligned}$$

Thus, if $(n+1)a+l$ is odd, then $P_{l,n}(1) = 0$. □

Lemma 3

We have that

$$\lim_{n \rightarrow \infty} |S_n(1)|^{1/n} = \varphi_{r,a}.$$

Furthermore, we have the inequalities

$$0 < \varphi_{r,a} \leq \frac{2^{r+1}}{r^{a-2r}}.$$

Proof. Let us write

$$\tilde{R}_n(k) = n!^{a-2r} \frac{(k - rn + 1)_{rn} (k + n + 2)_{rn}}{(k + 2)_n^a}$$

so that $R_n(k) = \frac{1}{(k+1)^a} \tilde{R}_n(k)$. We define an auxiliary function $F : (r, \infty) \rightarrow \mathbb{R}$ by

$$F(x) = \frac{x^{x(a+1)} (x + r + 1)^{x+r+1}}{(x+1)^{(x+1)(a+1)} (x-r)^{x-r}}.$$

We divide the proof into steps:

1. The function F admits a maximum at $x_0 = \frac{s_0}{1-s_0}$ and $F(x_0) = \varphi_{r,a}$.
2. There exists $c = c(a, r) > 0$ (independent of n) such that

$$\max_{k \geq 0} \tilde{R}_n(k) = \max_{rn \leq k \leq cn} \tilde{R}_n(k)$$

3. By letting M_n be the maximum of $\tilde{R}_n(k)$, we have:

$$\frac{1}{(cn)^a} M_n \leq S_n(1) \leq M_n$$

for $n \geq 1$.

4. If we write

$$\tilde{R}_n(k) = \rho_n(k) \frac{k^{k(a+1)} (k + (r+1)n)^{k+(r+1)n} \cdot n^{n(a-2r)}}{(k+n)^{(k+n)(a+1)} (k-rn)^{k-rn}}$$

then $\lim_{n \rightarrow \infty} \rho_n(k)^{1/n} = 1$ for $rn \leq k \leq cn$. Here, $k = k(n)$ denotes a function of n satisfying $r \leq k(n)/n \leq c$. This in particular implies that $k(n) \rightarrow \infty$.

5. We have $\lim_{n \rightarrow \infty} M_n^{1/n} = \varphi_{r,a}$.

Using the previous steps, we can take absolute values of the third inequality, then elevate to the $1/n$ power, and use the last point to conclude the result. Let us prove every point:

1. If $G(x) = \log F(x)$, we obtain:

$$\begin{aligned} G'(x) &= -(a+1) \log(x+1) + (a+1) \log(x) + \log(r+x+1) - \log(-r+x) \\ &= \log \left(\frac{x^{a+1} \cdot (x+r+1)}{(x-r) \cdot (x+1)^{a+1}} \right) \end{aligned}$$

If $G'(x) = 0$, then the term inside the logarithm is equal to 1. If we substitute $s = \frac{x}{x+1}$ in $Q_{a,r}(X)$, we show that:

$$Q_{r,a}(s) = rs^{a+2} - (r+1)s^{a+1} + (r+1)s - r = 0$$

Since $x > 0$, then $s \in (0, 1)$, and so $s = s_0$, where s_0 is the unique root of Q dans $(0, 1)$. We deduce that $\frac{x}{x+1} = s_0$, so $x = \frac{s_0}{1-s_0}$. If we let $x_0 = \frac{s_0}{1-s_0}$, a similar reasoning using our previous analysis of Q allows to show that G attains a maximum in x_0 .

In order to establish

$$F(x_0) = \varphi_{r,a} = ((r+1)s_0 - r)^r ((r+1) - rs_0)^{r+1} (1 - s_0)^{a-2r}$$

we obtain the relation $G'(x_0) = 0$, which is equivalent to

$$\frac{x_0^{a+1} \cdot (x_0 + r + 1)}{(x_0 - r) \cdot (x_0 + 1)^{a+1}} = 1$$

and therefore

$$x_0 + r + 1 = \frac{(x_0 - r) \cdot (x_0 + 1)^{a+1}}{x_0^{a+1}}.$$

By substituting this expression in the definition of $F(x_0)$, we obtain:

$$F(x_0) = \frac{(x_0 - r)^{2r+1} (x_0 + 1)^{r(a+1)}}{x_0^{(a+1)(r+1)}}$$

Finally, substituting $x_0 = \frac{s_0}{1-s_0}$ gives us $F(x_0) = \varphi_{r,a}$, as needed.

2. Since $\tilde{R}_0 = 1$, we can assume $n \geq 1$. Note that $\tilde{R}_n(k) = 0$ for $k < rn$, which means the maximum is attained when $rn \leq k$. In order to prove we can restrict ourselves to indices satisfying $k \leq cn$, we introduce the variable $x = k/n$. In this way, we have to establish the existence of a constant $c = c(a, r) > 0$ such that for all $x \geq c$:

$$\frac{\tilde{R}_n((x+1)n)}{\tilde{R}_n(xn)} < 1.$$

This property ensures that $\tilde{R}_n(xn)$ is eventually decreasing, hence:

$$\max_{k \geq 0} \tilde{R}_n(k) = \max_{rn \leq k} \tilde{R}_n(k) = \max_{r \leq x} \tilde{R}_n(xn) = \max_{r \leq x \leq c} \tilde{R}_n(xn) = \max_{rn \leq k \leq cn} \tilde{R}_n(k).$$

A direct calculation shows that:

$$\begin{aligned} \frac{\tilde{R}_n((x+1)n)}{\tilde{R}_n(xn)} &= \prod_{l=0}^{rn-1} \left(1 + \frac{n}{xn - rn + 1 + l}\right) \left(1 + \frac{n}{xn + n + 2 + l}\right) \cdot \frac{1}{\prod_{l=0}^{n-1} \left(1 + \frac{n}{xn+2+l}\right)^a} \\ &= \prod_{l=0}^{rn-1} \left(1 + \frac{1}{x - r + (1+l)/n}\right) \left(1 + \frac{1}{x + 1 + (2+l)/n}\right) \cdot \frac{1}{\prod_{l=0}^{n-1} \left(1 + \frac{1}{x+(2+l)/n}\right)^a} \\ &\leq \prod_{l=0}^{rn-1} \left(1 + \frac{1}{x-r}\right) \left(1 + \frac{1}{x+1}\right) \frac{1}{\prod_{l=0}^{n-1} \left(1 + \frac{1}{x+2}\right)^a} \\ &= \left(\left(1 + \frac{1}{x-r}\right)^r \left(1 + \frac{1}{x+1}\right)^r \left(1 + \frac{1}{x+2}\right)^{-a} \right)^n \\ &= \left(\left(1 + \frac{r}{x-r} + o(1/x)\right) \left(1 + \frac{r}{x+1} + o(1/x)\right) \left(1 - \frac{a}{x+2} + o(1/x)\right) \right)^n \\ &= \left(1 + \frac{r}{x-r} + \frac{r}{x+1} - \frac{a}{x+2} + o(1/x) \right)^n \\ &= \left(1 - \frac{a-2r}{x} + o(1/x) \right)^n. \end{aligned}$$

Since $a - 2r > 0$, and since the $o(\cdot)$ are independent of n , the quantity inside the parentheses is eventually less than 1, hence the whole expression is strictly less than 1, meaning that there is some integer c independent of n such that for $x \geq c$:

$$\frac{\tilde{R}_n((x+1)n)}{\tilde{R}_n(xn)} < 1$$

Without loss of generality, we can suppose $c > x_0$, so that the maximum is attained for $rn \leq k < cn$. In conclusion,

$$M_n = \max_{rn \leq k < cn} \tilde{R}_n(k) \quad \text{et} \quad \varphi_{r,a} = \max_{x \in (r, \infty)} F(x) = \max_{x \in (r, c]} F(x).$$

3. We know

$$\sum_{k \geq rn} \frac{1}{(k+1)^a} \leq \frac{1}{2^3} + \frac{1}{3^3} + \dots < 1$$

hence

$$S_n(1) = \sum_{k \geq 0} R_n(k) = \sum_{k \geq rn} R_n(k) = \sum_{k \geq rn} \frac{1}{(k+1)^a} \tilde{R}_n(k) < M_n.$$

For the other inequality, we consider k_0 such that $rn \leq k_0 < cn$ and such that $R_n(k_0) = M_n$. Since $\tilde{R}_n(k) > 0$ for every $k \geq rn$, then:

$$S_n(1) = \sum_{k \geq rn} \frac{1}{(k+1)^a} \tilde{R}_n(k) \geq \frac{1}{(k_0+1)^a} M_n \geq \frac{1}{(cn)^a} M_n.$$

4. We have

$$\rho_n(k) = n!^{a-2r} \frac{k!}{(k-rn)!} \frac{(k+(r+1)n+1)!}{(k+n+1)!} \frac{(k+n)^{(k+n)(a+1)}(k-rn)^{k-rn}}{\frac{(k+n+1)!^a}{(k+1)!^a} k^{k(a+1)}(k+(r+1)n)^{k+(r+1)n} \cdot n^{n(a-2r)}}.$$

By Stirling's formula, $(n!)^{1/n} \sim n/e$. Since $k = k(n)$ tends to infinity when n tends to infinity, we also have $(k!)^{1/k} \sim k/e$, thus $(k!)^{1/n} = ((k!)^{1/k})^{k/n} \sim (k/e)^{k/n}$. We can elevate both sides of the equivalence to the k/n -th power because $r \leq k/n \leq c$. By using the same reasoning to the other factorials, we conclude that $\rho_n(k)^{1/n}$ is equivalent to

$$\frac{\left(\frac{n}{e}\right)^{a-2r} \left(\frac{k}{e}\right)^{k/n} \left(\frac{k+(r+1)n+1}{e}\right)^{\frac{k+(r+1)n+1}{n}} \left(\frac{k+1}{e}\right)^{a \frac{k+1}{n}}}{\left(\frac{k-rn}{e}\right)^{\frac{k-rn}{n}} \left(\frac{k+n+1}{e}\right)^{\frac{k+n+1}{n}} \left(\frac{k+n+1}{e}\right)^{a \frac{k+n+1}{n}}} \frac{(k+n)^{\frac{(k+n)(a+1)}{n}} (k-rn)^{\frac{k-rn}{n}}}{k^{\frac{k(a+1)}{n}} (k+(r+1)n)^{\frac{k+(r+1)n}{n}} n^{a-2r}}.$$

After simplification, the exponent of e turns out to be 0, and the term n^{a-2r} disappears. By regrouping similar terms, we can show that the whole expression tends to 1, for example:

$$\frac{k^{k/n} (k+1)^{a \frac{k+1}{n}}}{k^{\frac{k(a+1)}{n}}} = \left(\frac{(k+1)^{a(k+1)}}{k^{ka}} \right)^{1/n} = \left(\left(1 + \frac{1}{k} \right)^{ak} \right)^{1/n} (k+1)^{a/n}$$

Since $k(n) \rightarrow \infty$, the first term inside the parentheses tends to $\exp(a) > 0$, and so the $1/n$ -th power tends to 1. The second term tends to 1 because $rn \leq k \leq cn$. Similar arguments show that the whole product tends to 1.

5. Finally, let us show that $\varphi_{r,a} = \lim_{n \rightarrow \infty} M_n^{1/n}$. The functions ρ_n and F were defined so that:

$$F(k/n) = \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n}$$

By continuity of F , we have:

$$\varphi_{r,a} = \max_{x \in [r,c]} F(x) = \lim_{n \rightarrow \infty} \max_{rn \leq k \leq cn} F(k/n) = \lim_{n \rightarrow \infty} \max_{rn \leq k \leq cn} \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n}.$$

Additionally, we have showed that for every $\varepsilon > 0$ there is some N such that for $n \geq N$ and for each k with $rn \leq k \leq cn$:

$$1 - \varepsilon < \frac{1}{\rho_n(k)^{1/n}} < 1 + \varepsilon$$

Technically, we formulated this using a function $k = k(n)$ and with the fact that $\lim_{n \rightarrow \infty} \rho_n(k(n))^{1/n} \rightarrow 1$, but we can easily show the equivalence between this formulation and the one that we just stated.

We deduce:

$$\begin{aligned} (1 - \varepsilon)M_n^{1/n} &= (1 - \varepsilon) \max_{rn \leq k \leq cn} \tilde{R}_n(k)^{1/n} \\ &\leq \max_{rn \leq k \leq cn} \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n} \\ &\leq (1 + \varepsilon) \max_{rn \leq k \leq cn} \tilde{R}_n(k) = (1 + \varepsilon)M_n^{1/n} \end{aligned}$$

and then:

$$1 - \varepsilon \leq \frac{1}{M_n^{1/n}} \max_{rn \leq k \leq cn} \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n} \leq 1 + \varepsilon.$$

We have established that:

$$\lim_{n \rightarrow \infty} \frac{1}{M_n^{1/n}} \max_{rn \leq k \leq cn} \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n} = 1$$

and thus:

$$\varphi_{r,a} = \lim_{n \rightarrow \infty} \max_{rn \leq k \leq cn} \left(\frac{\tilde{R}_n(k)}{\rho_n(k)} \right)^{1/n} = \lim_{n \rightarrow \infty} M_n^{1/n}.$$

Lastly, we need to prove the bound $0 < \varphi_{r,a} < 2^{r+1}/r^{a-2r}$. Since $\varphi_{r,a} = \max_x F(x)$ and $F > 0$, we immediately obtain $\varphi > 0$. For the other bound, if we admit the inequality $k + (r+1)n < 2^{1+1/r}k$ for $k > rn$, we can bound each Pochhammer symbol of $\tilde{R}_n(k)$ to get:

$$\tilde{R}_n(k) < n^{(a-2r)} \frac{k^{rn} (2^{1+1/r}k)^{rn}}{k^{an}} = \left(2^{r+1} \left(\frac{n}{k} \right)^{a-2r} \right)^n < \left(\frac{2^{r+1}}{r^{a-2r}} \right)^n.$$

Hence:

$$\varphi_{r,a} = \lim_{n \rightarrow \infty} M_n^{1/n} = \lim_{n \rightarrow \infty} \left(\max_{rn \leq k \leq cn} \tilde{R}_n(k) \right)^{1/n} \leq \frac{2^{r+1}}{r^{a-2r}}$$

which is what we wanted. To prove the inequality $2^{1+1/r}k \geq k + (r+1)n$, we use that

$$2^x \geq 1 + x \log 2 \geq 1 + x/2$$

and therefore $2^{1+x} \geq 2 + x$. By applying this property to $x = 1/r$, we obtain:

$$\begin{aligned} k + (r+1)n &< k + \frac{r+1}{r}k \\ &= k \cdot \left(2 + \frac{1}{r}\right) \\ &\leq k2^{1+1/r} \end{aligned}$$

which finishes the proof. □

Lemma 4

For every $l \in \llbracket 0, a \rrbracket$,

$$\limsup_{n \rightarrow \infty} |P_{l,n}(1)|^{1/n} \leq 2^{a-2r} (2r+1)^{2r+1}.$$

Proof. We divide the proof in two cases: First, we consider the case $l \in \llbracket 1, a \rrbracket$. In this case,

$$P_{l,n}(1) = \sum_{j=0}^n c_{l,j,n}.$$

To bound $|P_{l,n}(1)|^{1/n}$, it suffices to bound the coefficients $|c_{l,j,n}|$. Since

$$R_n(z) = \sum_{l=1}^a \sum_{j=0}^n \frac{c_{l,j,n}}{(z+j+1)^l}$$

then, for a fixed $j \in \llbracket 0, n \rrbracket$, we have

$$R_n(z)(z+j+1)^{l-1} = Q_n(z) + \frac{c_{l,j,n}}{(z+j+1)}$$

where Q_n is the sum of meromorphic functions with denominators different than $z+j+1$. By using Cauchy's formula with the function $z \mapsto c_{l,j,n}$ we obtain:

$$\begin{aligned} c_{l,j,n} &= \frac{1}{2\pi i} \int_{|z+j+1|=1/2} \frac{c_{l,j,n}}{z+j+1} dz \\ &= \frac{1}{2\pi i} \int_{|z+j+1|=1/2} \left(Q_n(z) + \frac{c_{l,j,n}}{z+j+1} \right) dz \\ &= \frac{1}{2\pi i} \int_{|z+j+1|=1/2} R_n(z)(z+j+1)^{l-1} dz. \end{aligned}$$

Since

$$R_n(z) = n!^{a-2r} \frac{(z-rn+1)_{rn}(z+n+2)_{rn}}{(z+1)_{n+1}^a},$$

we can bound the quantities that appear in the product when $|z + j + 1| = 1/2$ in order to bound $c_{l,j,n}$. Using the triangle inequality,

$$\begin{aligned}
|(z - rn + 1)_{rn}| &= \prod_{i=0}^{rn-1} |z - rn + 1 + i| \\
&= \prod_{i=0}^{rn-1} |z + j + 1 - j - rn + i| \\
&\leq \prod_{i=0}^{rn-1} \left(\frac{1}{2} + |rn - i + j| \right) \\
&= \prod_{i=0}^{rn-1} \left(\frac{1}{2} + rn - i + j \right) \\
&\leq \prod_{i=0}^{rn-1} (1 + rn - i + j) \\
&= \prod_{i=0}^{rn-1} (j + 2 + (rn - i - 1)) \\
&= (j + 2)_{rn}.
\end{aligned}$$

Similarly

$$\begin{aligned}
|(z + n + 2)_{rn}| &= \prod_{i=0}^{rn-1} (z + n + 2 + i) \\
&= \prod_{i=0}^{rn-1} |z + j + 1 + n + 1 + i - j| \\
&\leq \prod_{i=0}^{rn-1} \left(\frac{1}{2} + |n + 1 + i - j| \right) \\
&= \prod_{i=0}^{rn-1} \left(\frac{1}{2} + n + 1 + i - j \right) \\
&\leq \prod_{i=0}^{rn-1} (n - j + 2 + i) \\
&= (n - j + 2)_{rn}.
\end{aligned}$$

Finally, we bound $|(z+1)_{n+1}|$ from below, by bounding each $|z+1+i|$ for $0 \leq i \leq n$. Let us note that

$$|z+j+1| = \frac{1}{2}$$

$$|z+j| \geq |1 - |z+j+1|| = \frac{1}{2}$$

$$|z+j+2| \geq |1 - |z+j+1|| = \frac{1}{2}.$$

Therefore,

$$|(z+1)_{n+1}| \geq \frac{1}{8} \prod_{i=0}^{j-2} |z+1+i| \prod_{i=j+2}^n |z+1+i|.$$

For every $i \in \llbracket 0, n \rrbracket$,

$$|z+1+i| = |z+j+1+i-j| \geq \left| |i-j| - \frac{1}{2} \right| \geq |i-j| - 1$$

and therefore

$$\prod_{i=0}^{j-2} |z+1+i| \geq \prod_{i=0}^{j-2} (j-i-1) = (j-1)!$$

$$\prod_{i=j+2}^n |z+1+i| \geq \prod_{i=j+2}^n (i-j-1) = (n-j-1)!.$$

We have concluded that

$$|(z+1)_{n+1}| \geq \frac{1}{8} (j-1)! (n-j-1)!.$$

By using every proven bound,

$$|c_{l,j,n}| \leq \frac{1}{2\pi} \int_{|z+j+1|=\frac{1}{2}} n!^{a-2r} \frac{(j+2)_{rn} (n-j+2)_{rn}}{\left(\frac{1}{8}(j-1)!(n-j-1)!\right)^a} \left(\frac{1}{2}\right)^{l-1} dz.$$

The expression inside of the integral does not depend on z . Using the bound $\frac{1}{2^{l-1}} \leq 2$, we obtain:

$$\begin{aligned} |c_{l,j,n}| &\leq n!^{a-2r} \frac{(j+2)_{rn} (n-j+2)_{rn}}{\left(\frac{1}{8}(j-1)!(n-j-1)!\right)^a} \\ &= n!^{a-2r} \frac{(j+rn+1)!}{(j+1)!} \frac{((r+1)n-j+1)!}{(n-j+1)!} \frac{8^a}{((j-1)!(n-j-1)!)^a} \\ &= \frac{(j+rn+1)!}{(j+1)!(j!(n-j)!)^r} \frac{((r+1)n-j+1)!}{(n-j+1)!(j!(n-j)!)^r} \left(\frac{n!}{j!(n-j)!}\right)^{a-2r} \cdot \frac{(8j(n-j))^a}{((j-1)!(n-j-1)!)^a} \\ &\leq \frac{(j+rn+1)!}{(j+1)!(j!(n-j)!)^r} \frac{((r+1)n-j+1)!}{(n-j+1)!(j!(n-j)!)^r} \left(\frac{n!}{j!(n-j)!}\right)^{a-2r} \cdot (8j(n-j))^a \end{aligned}$$

Let us note that the terms

$$\frac{(j + rn + 1)!}{(j + 1)!(j!(n - j)!)^r} \quad \text{et} \quad \frac{((r + 1)n - j + 1)!}{(n - j + 1)!(j!(n - j)!)^r}$$

are multinomial coefficients. We know that

$$\begin{aligned} (2r + 1)^{rn+j+1} &= \underbrace{(1 + \dots + 1)}_{2r+1}^{rn+j+1} \\ &= \sum_{x_1+x_2+\dots+x_{2r+1}=rn+j+1} \frac{(rn + j + 1)!}{x_1!x_2! \dots x_{2r+1}!} \\ &\geq \frac{(j + rn + 1)!}{(j + 1)!(j!(n - j)!)^r}. \end{aligned}$$

And similarly, we can show

$$\frac{((r + 1)n - j + 1)!}{(n - j + 1)!(j!(n - j)!)^r} \leq (2r + 1)^{(r+1)n-j+1}.$$

We bound the term $\binom{n}{j}$ with 2^n . Additionally, using $xy \leq (\frac{x+y}{2})^2$ with $x = j$ and $y = n - j$, we get

$$8j(n - j) \leq 8 \left(\frac{n}{2} \right)^2 = 2n^2.$$

Finally, putting together every bound allows us to conclude

$$|c_{l,j,n}| \leq (2r + 1)^{(2r+1)n+2} 2^{(a-2r)n} (2n^2)^a.$$

Therefore

$$\begin{aligned} \limsup_{n \rightarrow \infty} |P_{l,n}(1)|^{1/n} &\leq \limsup_{n \rightarrow \infty} (n + 1)^{1/n} \max_{j \in \llbracket 0, n \rrbracket} |c_{l,j,n}|^{1/n} \\ &\leq \limsup_{n \rightarrow \infty} (2r + 1)^{2r+1+2/n} 2^{a-2r} \underbrace{(2n^2)^{a/n} n^{1/n}}_{\rightarrow 1} \\ &= (2r + 1)^{2r+1} 2^{a-2r} \end{aligned}$$

as we wanted to prove.

It remains to bound $|P_{0,n}(1)|$, with

$$P_{0,n}(1) = \sum_{l=1}^a \sum_{j=1}^n \sum_{k=0}^{j-1} \frac{-c_{l,j,n}}{(k + 1)^l}$$

For each $(l, j) \in \llbracket 1, a \rrbracket \times \llbracket 1, n \rrbracket$, we know that

$$\sum_{k=0}^{j-1} \frac{1}{(k + 1)^l} \leq \sum_{k=0}^{j-1} \frac{1}{(k + 1)} \leq j \leq n$$

and so, similar to the previous case,

$$\begin{aligned} \limsup_{n \rightarrow \infty} |P_{0,n}(1)|^{1/n} &\leq \limsup_{n \rightarrow \infty} (an^2)^{1/n} \max_{j \in \llbracket 0, n \rrbracket} |c_{l,j,n}|^{1/n} \\ &\leq (2r+1)^{2r+1} 2^{a-2r}. \end{aligned}$$

□

Lemma 5

For $n \in \mathbb{N}^*$, we let $d_n = \text{lcm}(1, \dots, n)$. Hence, for each $l \in \llbracket 0, a \rrbracket$,

$$d_n^{a-l} P_{l,n}(z) \in \mathbb{Z}[z].$$

Proof. It suffices to prove that $d_n^{a-l} P_{l,n}(z) \in \mathbb{Z}[z]$ for every $l \in \llbracket 0, a \rrbracket$ and $j \in \llbracket 0, n \rrbracket$. Let $(l, j) \in \llbracket 0, a \rrbracket \times \llbracket 0, n \rrbracket$. Recall that

$$R_n(z) = n!^{a-2r} \frac{(z - rn + 1)_{rn} (z + n + 2)_{rn}}{(z + 1)_{n+1}^a}.$$

If we define, for each $k \in \llbracket 1, r \rrbracket$

$$F_k(z) = \frac{(z - kn + 1)_n}{(z + 1)_{n+1}} (z + j + 1)$$

$$G_k(z) = \frac{(z + kn + 2)_n}{(z + 1)_{n+1}} (z + j + 1)$$

$$H(z) = \frac{n!}{(z + 1)_{n+1}} (z + j + 1),$$

we can write

$$R_n(z)(z + j + 1)^a = \prod_{k=1}^r F_k(z) \times \prod_{k=1}^r G_k(z) \times H(z)^{a-2r}.$$

This is just writing the Pochhammer symbols in r products of size n , for example:

$$(z - rn + 1)_{rn} = \prod_{k=1}^r (z - kn + 1)_n.$$

If we fix $k \in \llbracket 1, r \rrbracket$:

$$\begin{aligned} F_k(z) &= \frac{(z - kn + 1)_n}{(z + 1) \cdots (z + n + 1)} (z + j + 1) \\ &= \frac{(z - kn + 1)_n}{\prod_{\substack{s=0 \\ s \neq j}}^n (z + s + 1)} \end{aligned}$$

then F_k is a rational function of zero degree. Its poles are in $-(s+1) < 0$ for $s \in \llbracket 0, n \rrbracket \setminus \{j\}$, since the terms in the numerator are equal to zero when $z = kn - s \geq 0$ for $s \in \llbracket 1, n \rrbracket$. We also remark that the poles have order 1.

By the partial fraction decomposition theorem, there is some $C_k \in \mathbb{Q}$ and $(C_{k,s})_s \in \mathbb{Q}^n$ such that

$$F_k(z) = C_k + \sum_{\substack{s \neq j \\ s=0}}^n \frac{C_{k,s}}{z + s + 1}.$$

Taking the limit of this expression when $z \in \mathbb{R}$ and $z \rightarrow \infty$, and using that $\lim_{z \rightarrow \infty} F_k(z) = 1$, we get $C_k = 1$.

Additionally, we can compute the $C_{k,s}$ as the residues of $F_k(s)$ in $-(s+1)$:

$$\begin{aligned} C_{k,s} &= \text{Res}(F_k, -(s+1)) \\ &= \lim_{z \rightarrow -(s+1)} (z + s + 1) F_k(z) \\ &= \frac{(-(s+1) - kn + 1)_n}{\prod_{\substack{t=0 \\ t \notin \{j, s\}}}^n (-(s+1) + t + 1)} \\ &= \frac{(-s - kn)_n}{\prod_{\substack{t=0 \\ t \neq s}}^n (t - s)} \cdot (j - s) \\ &= \frac{\prod_{i=0}^{n-1} (-s - kn + i)}{\prod_{t=0}^{s-1} (t - s) \cdot \prod_{t=s+1}^n (t - s)} \cdot (j - s) \\ &= \frac{(-1)^n \prod_{i=0}^{n-1} (s + kn - i)}{(-1)^s s! (n - s)!} \cdot (j - s) \\ &= (-1)^{n-s} \frac{(s + kn)!}{(s + (k-1)n)! n!} \frac{n!}{s! (n - s)!} \cdot (j - s) \\ &= \underbrace{(-1)^{n-s} \binom{s + kn}{n} \binom{n}{s}}_{f_{k,s} \in \mathbb{Z}} (j - s) \end{aligned}$$

We have shown that

$$F_k(z) = 1 + \sum_{\substack{s=0 \\ s \neq j}}^n \frac{(j - s) f_{k,s}}{z + s + 1}.$$

Similar arguments yield

$$G_k(z) = 1 + \sum_{\substack{s=0 \\ s \neq j}}^n \frac{(j-s)g_{k,s}}{z+s+1} \quad \text{and} \quad H(z) = \sum_{\substack{s=0 \\ s \neq j}}^n \frac{(j-s)h_s}{z+s+1}$$

with $g_{k,s}, h_s \in \mathbb{Z}$ for every $s \in \llbracket 0, n \rrbracket \setminus \{j\}$. Having done this, we can compute the derivatives of F_k , G_k and H of order $\lambda \geq 0$:

$$D_\lambda F_l(z)|_{z=-(j+1)} = \delta_{0,\lambda} + \sum_{\substack{s=0 \\ s \neq j}}^n (-1)^\lambda \frac{(j-s)f_{s,l}}{(s-j)^{\lambda+1}}$$

$$= \delta_{0,\lambda} + \sum_{\substack{s=0 \\ s \neq j}}^n (-1)^{\lambda+1} \frac{f_{s,l}}{(s-j)^\lambda}$$

$$D_\lambda G_l(z)|_{z=-(j+1)} = \delta_{0,\lambda} + \sum_{\substack{s=0 \\ s \neq j}}^n (-1)^\lambda \frac{g_{s,l}}{(s-j)^\lambda}$$

$$D_\lambda H(z)|_{z=-(j+1)} = \sum_{\substack{s=0 \\ s \neq j}}^n (-1)^\lambda \frac{h_s}{(s-j)^\lambda}.$$

with $\delta_{0,\lambda}$ equal to 1 if $\lambda = 0$ and 0 if not.

We remark that the denominators of the previous quantities are numbers which belong to $\llbracket 1, n \rrbracket$, taken to the λ -th power. Therefore,

$$d_n^\lambda D_\lambda F_k(z)|_{z=-(j+1)} \quad d_n^\lambda D_\lambda G_k(z)|_{z=-(j+1)} \quad d_n^\lambda D_\lambda H(z)|_{z=-(j+1)}$$

are in $\mathbb{Z}$ for every $\lambda \geq 0$.

Doing this for every $k \in \llbracket 1, r \rrbracket$, Leibniz's formula yields:

$$\begin{aligned} d_n^{a-l} c_{l,j,n} &= d_n^{a-l} D_{a-l}((R(z))(z+j+1)^a) \\ &= d_n^{a-l} D_{a-l} \left(\prod_{k=1}^r F_k(z) \times \prod_{k=1}^r G_k(z) \times H(z)^{a-2r} \right) \\ &= \sum_{\mu_1 + \dots + \mu_a = a-l} (d_n^{\mu_1} D_{\mu_1} F_1) \dots (d_n^{\mu_r} D_{\mu_r} F_r) \times \\ &\quad (d_n^{\mu_{r+1}} D_{\mu_{r+1}} G_1) \dots (d_n^{\mu_{2r}} D_{\mu_{2r}} G_r) \times \\ &\quad (d_n^{\mu_{2r+1}} D_{\mu_{2r+1}} H) \dots (d_n^{\mu_a} D_{\mu_a} H) \end{aligned}$$

This implies $d_n^{a-l} c_{l,j,n} \in \mathbb{Z}$, and so we conclude $d_n^{a-l} P_{l,n}(z) \in \mathbb{Z}[z]$. □

5 The main results

The next proposition will help us prove our two main theorems:

Proposition 1

Let $a \geq 3$ be an odd integer. For every r such that $1 \leq r < a/2$ we have the lower bound

$$\delta(a) \geq \frac{(a-2r)\log 2 + (2r+1)\log(2r+1) - \log(\varphi_{r,a})}{a + (a-2r)\log(2) + (2r+1)\log(2r+1)}.$$

In particular,

$$\delta(a) \geq \frac{\log(r) + \frac{a-r}{a+1}\log(2)}{1 + \log(2) + \frac{2r+1}{a+1}\log(r+1)}$$

Proof. Observe that another way to write $d_n = \text{lcm}(1, \dots, n)$ is

$$d_n = \prod_{p \text{ prime}} p^{\lfloor \log_p n \rfloor}.$$

This means

$$\begin{aligned} \log d_n &= \sum_{p \text{ prime}} \lfloor \log_p n \rfloor \cdot \log(p) \\ &= \sum_{p \text{ prime}} \sum_{p^k \leq n} \log p \\ &= \sum_{p^k \leq n} \log p = \psi(n) \end{aligned}$$

where ψ is the Tchebychev function. By the prime number theorem, $\psi(n) = n + o(n)$, and thus $d_n = e^{n+o(n)}$. Let us define $n \geq 0$ the number $\ell_n = d_{2n}^a S_{2n}(1)$ and

$$p_{0,n} = d_{2n}^a P_{0,2n}(1)$$

$$p_{k,n} = d_{2n}^a P_{2k+1,2n}(1) \quad \text{for } k \in \left[1, \frac{a-1}{2}\right].$$

If we multiply the second equation of [Lemma 2](#) by d_{2n}^a we obtain

$$\ell_n = p_{0,n} + \sum_{k=1}^{\frac{a-1}{2}} p_{k,n} \zeta(2k+1).$$

Thanks to [Lemma 5](#), we get $p_{0,n} \in \mathbb{Z}$. For $k \in \left[1, \frac{a-1}{2}\right]$, we have $d_{2n}^{a-(2k+1)} P_{2k+1,2n}(1) \in \mathbb{Z}$, and by multiplying by $d_{2n}^{2k+1} \in \mathbb{Z}$, we conclude $p_{k,n} \in \mathbb{Z}$.

It remains to show the second and third conditions of Nesterenko's criterion. Let $\alpha = (e^a \varphi_{r,a})^2$ and $\beta = (e^a 2^{a-2r} (2r+1)^{2r+1})^2$. Evidently $\beta > 1$, so it suffices to show the inequalities

$$|\ell_n| = \alpha^{n+o(n)} \quad \text{et} \quad |p_{k,n}| \leq \beta^{n+o(n)}$$

which are equivalent to

$$\log |\ell_n| = n \log \alpha + o(n) \quad \text{and} \quad \log |p_{k,n}| \leq n \log \beta + o(n).$$

First, let us compute $\log |\ell_n|$. By definition $\ell_n = d_{2n}^a S_{2n}(1)$, so

$$\log |\ell_n| = a \log(d_{2n}) + \log |S_{2n}(1)|.$$

We know that $a \log(d_{2n}) = a(2n + o(2n)) = 2an + o(n)$. We also know by [Lemma 3](#) that

$$\lim_{n \rightarrow \infty} |S_{2n}(1)|^{\frac{1}{2n}} = \varphi_{r,a} \quad \text{and} \quad \varphi_{r,a} \neq 0$$

which means

$$\frac{|S_{2n}(1)|^{\frac{1}{2n}}}{\varphi_{r,a}} = 1 + o(1)$$

and thus

$$\frac{1}{2n} \log |S_{2n}(1)| - \log \varphi_{r,a} = \log(1 + o(1)) = o(1).$$

Consequently, we obtain $\log |S_{2n}(1)| = 2n \log \varphi_{r,a} + o(n)$. Putting together every bound yields

$$\begin{aligned} \log |\ell_n| &= 2an + o(n) + 2n \log \varphi_{r,a} + o(n) \\ &= 2n(\log(e^a \varphi_{r,a})) + o(n) \\ &= n \log \alpha + o(n) \end{aligned}$$

We have shown the second condition of Nesterenko's criterion. Now, for $k \in \llbracket 0, \frac{a-1}{2} \rrbracket$, let us show that $\log |p_{k,n}| \leq n \log \beta + o(n)$. Thanks to [Lemma 4](#), we know that for every $l \in \llbracket 0, a \rrbracket$,

$$\limsup_{n \rightarrow \infty} |P_{l,2n}(1)|^{1/2n} \leq 2^{a-2r} (2r+1)^{2r+1}.$$

Let $C = 2^{a-2r} (2r+1)^{2r+1}$. Using the convention that $\log$ is equal to $-\infty$ when evaluated at $-\infty$, we can use the fact that $\log$ is continuous and increasing to obtain

$$\limsup_{n \rightarrow \infty} \frac{1}{2n} \log |P_{l,2n}(1)| \leq \log(C)$$

For $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for $n \geq N$,

$$\frac{1}{2n} \log |P_{l,2n}(1)| < \log(C) + \varepsilon,$$

and therefore if we define

$$r_n = \max \left\{ 0, \frac{1}{2n} \log |P_{l,2n}(1)| - \log(C) \right\}.$$

we will have $0 \leq r_n < \varepsilon$ for $n \geq N$, which shows that $\lim_{n \rightarrow \infty} r_n = 0$. We know

$$\frac{1}{2n} \log |P_{l,2n}(1)| - \log(C) \leq r_n$$

so

$$\log |P_{l,2n}(1)| \leq 2n \log(C) + 2nr_n$$

which shows $\log |P_{l,2n}(1)| \leq 2n \log(C) + o(n)$. Once again, for $k \in \llbracket 1, \frac{a-1}{2} \rrbracket$ we have $p_{k,n} = d_{2n}^a P_{2k+1,2n}(1)$, so we bring together every bound to conclude

$$\begin{aligned} \log |p_{k,n}| &= a \log d_{2n} + \log |P_{2k+1,2n}(1)| \\ &\leq 2an + o(n) + 2n \log C + o(n) \\ &= 2n \left(\log(e^a 2^{a-2r} (2r+1)^{2r+1}) \right) + o(n) \\ &= n \log \beta + o(n). \end{aligned}$$

The same type of argument works for $p_{0,n} = d_{2n}^a P_{0,2n}(1)$. So, we can finally apply Nesterenko's criterion, and some simplification, to obtain

$$\delta(a) \geq \frac{\log \beta - \log \alpha}{\log \beta} = \frac{(a-2r) \log(2) + (2r+1) \log(2r+1) - \log(\varphi_{r,a})}{a + (a-2r) \log(2) + (2r+1) \log(2r+1)}$$

For the second claim of the proposition, we use the bound $\varphi_{r,a} \leq \frac{2^{r+1}}{r^{a-2r}}$ from [Lemma 3](#) and the bounds $2(r+1) \geq 2r+1 \geq 2r$ to obtain

$$\begin{aligned} \delta(a) &\geq \frac{(a-2r) \log(2) + (2r+1)(\log(2) + \log(r)) - (r+1) \log(2) + (a-2r) \log(r)}{a + (a-2r) \log(2) + (2r+1)(\log(2) + \log(r+1))} \\ &= \frac{(a+1) \log r + (a-r) \log(2)}{a + (a+1) \log(2) + r(2r+1) \log(r+1)} \\ &\geq \frac{(a+1) \log r + (a-r) \log(2)}{(a+1) + (a+1) \log(2) + (2r+1) \log(r+1)} \\ &= \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{1 + \log(2) + \frac{2r+1}{a+1} \log(r+1)} \end{aligned}$$

□

Now, we are ready to prove our two main theorems. We restate for reference:

Theorem

There exists an odd integer $j \leq 169$ such that $1, \zeta(3), \zeta(j)$ are linearly independent over $\mathbb{Q}$.

Proof. We choose $a = 169$ and $r = 10$. We can compute approximate s_0 with a computer and therefore also approximate $\varphi_{10,169}$. Using these computations and the first lower bound of [Proposition 1](#), we obtain $\delta(169) > 2,001$. Therefore, $\delta(169) \geq 3$, and since $\delta(3) = 2$, there exists a $3 < j \leq 169$ such that $1, \zeta(3), \zeta(j)$ are linearly independent over $\mathbb{Q}$. $\square$

Theorem

For $a \geq 3$ an odd integer, if we define $\delta(a) = \dim_{\mathbb{Q}} \text{Vect}_{\mathbb{Q}}\{1, \zeta(3), \dots, \zeta(a)\}$, then

$$\delta(a) \geq \frac{1}{3} \log(a).$$

Additionally, for every $\varepsilon > 0$, there is some $A(\varepsilon) \in \mathbb{N}$ such that if $a > A(\varepsilon)$, then

$$\delta(a) \geq \frac{1 - \varepsilon}{1 + \log(2)} \log(a).$$

We note that $\frac{1}{1+\log 2} \approx 0,5906$.

Proof. We split the proof by cases:

- If $3 \leq a \leq 167 < e^6$, then

$$\delta(a) \geq \delta(3) = 2 > \frac{1}{3} \log(a).$$

- If $169 \leq a \leq 8 \cdot 10^3 - 1 < e^9$, then [Theorem 2](#) gives us

$$\delta(a) \geq \delta(169) \geq 3 > \frac{1}{3} \log(a)$$

- If $8 \cdot 10^3 + 1 \leq a \leq 10^5 - 1 < e^{12}$, then using the second bound of [Proposition 1](#) with $r = 200$ gives us $\delta(8 \cdot 10^3 + 1) > 3$, and thus

$$\delta(a) \geq 4 \geq \frac{1}{3} \log(a).$$

- If $10^5 + 1 \leq a \leq 10^6 - 1 < e^{15}$, again the second bound of [Proposition 1](#) with $r = 600$ gives us $\delta(10^5 + 1) > 4$ and thus

$$\delta(a) \geq 5 \geq \frac{1}{3} \log(a)$$

- If $10^6 + 1 \leq a$, then we can choose $r = \lfloor a^{3/5} + 1 \rfloor$. Simple computations show that $1 \leq r < a/2$, and so we can apply [Proposition 1](#) to obtain

$$\begin{aligned}\delta(a) &\geq \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{1 + \log(2) + \frac{2r+1}{a+1} \log(r+1)} \\ &\geq \frac{\log(r)}{1 + \log(2) + \frac{2r+1}{a+1} \log(r+1)} \\ &\geq \frac{3}{5} \frac{\log(a)}{1 + \log(2) + \frac{2a^{3/5}+3}{a+1} \log(a^{3/5} + 2)}\end{aligned}$$

It can be shown that $g : a \mapsto 1 + \log(2) + \frac{2a^{3/5}+3}{a+1} \log(a^{3/5} + 2)$ is a decreasing function in $[1, \infty)$ and that $g(10^6 + 1) < 9/5$. Therefore

$$\delta(a) \geq \frac{1}{3} \log(a).$$

Let us show the second part of [Theorem 1](#). We define $r = r(a) = \lfloor \frac{a}{\log(a)^2} \rfloor$. We can suppose $a > 3$, so that simple calculations show $1 \leq r < a/2$. Therefore, we can apply [Proposition 1](#) to obtain:

$$\delta(a) \geq \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{1 + \log(2) + \frac{2r+1}{a+1} \log(r+1)}$$

If we define

$$H(a) = \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{1 + \log(2) + \frac{2r+1}{a+1} \log(r+1)} \cdot \frac{1 + \log 2}{\log a}$$

we will show that $\lim_{a \rightarrow \infty} H(a) = 1$, and therefore for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that for $a > N$,

$$\delta(a) \cdot \frac{1 + \log 2}{\log a} \geq H(a) \geq 1 - \varepsilon$$

which yields

$$\delta(a) \geq \frac{1 - \varepsilon}{1 + \log 2} \log a$$

which is the desired conclusion. For proving the limit, it suffices to show:

$$\lim_{a \rightarrow \infty} \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{\log a} = 1$$

$$\lim_{a \rightarrow \infty} \frac{\log(r) + \frac{a-r}{a+1} \log(2)}{1 + \log 2} = 1.$$

For the first one, since $r(a) \rightarrow \infty$ and $|r(a) - \frac{a}{\log(a)^2}| \leq 1$, then

$$\begin{aligned} \log(r) + \frac{a-r}{a+1} \log(2) &= \log\left(\frac{a}{\log(a)^2}\right) + \left(\log(r) - \log\left(\frac{a}{\log(a)^2}\right)\right) + \frac{a - \frac{a}{\log(a)^2} + O(1)}{a+1} \log(2) \\ &= \log(a) - 2 \log \log a + o(1) + (1 + o(1)) \log(2) \\ &= (1 + o(1)) \log(a) \end{aligned}$$

which implies what we wanted. For the denominator,

$$\begin{aligned} 0 &\leq \frac{2r+1}{a+1} \log(r+1) \leq \frac{2\frac{a}{\log(a)^2} + 1}{a+1} \log\left(\frac{a}{\log(a)^2} + 1\right) \\ &\leq \frac{3\frac{a}{\log(a)^2}}{a} \log\left(\frac{2a}{\log(a)^2}\right) \\ &= \frac{3 \log 2}{\log(a)^2} + \frac{3}{\log(a)} - \frac{6 \log(\log(a))}{\log(a)^2} \\ &= o(1) \end{aligned}$$

as we wanted to prove. □

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